

```
-- =====
-- Layer ۴: ERP Core Identity Layer (REVISED & SECURED)
-- =====
```

```
-- ۱. users
```

بر اساس اصل ایزوله‌سازی، ایندکس یکتای گلوبال از روی ایمیل و نام کاربری حذف شد --

```
CREATE TABLE users (
  user_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  mobile VARCHAR(۵۰),
  first_name VARCHAR(۱۰۰),
  last_name VARCHAR(۱۰۰),
  user_kind SMALLINT NOT NULL DEFAULT ۱,
  status SMALLINT NOT NULL DEFAULT ۱,
  last_login_at TIMESTAMPTZ,
  created_at TIMESTAMPTZ NOT NULL DEFAULT NOW(),
  created_by UUID,
  updated_at TIMESTAMPTZ,
  updated_by UUID,
  deleted_at TIMESTAMPTZ,
  deleted_by UUID,
  row_version BIGINT NOT NULL DEFAULT ۱
);
```

```
-- ۲. user_credentials
```

```
CREATE TABLE user_credentials (
  credential_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  user_id UUID NOT NULL REFERENCES users(user_id) ON DELETE RESTRICT, -- جهت امنیت داده RESTRICT تغییر به --
  password_hash VARCHAR(۵۰۰),
  authentication_type SMALLINT NOT NULL DEFAULT ۱,
  is_verified BOOLEAN NOT NULL DEFAULT FALSE,
  two_factor_enabled BOOLEAN NOT NULL DEFAULT FALSE,
  failed_login_count INT NOT NULL DEFAULT ۰,
  locked_until TIMESTAMPTZ,
  last_password_change_at TIMESTAMPTZ,
  created_at TIMESTAMPTZ NOT NULL DEFAULT NOW(),
  created_by UUID,
  updated_at TIMESTAMPTZ,
  updated_by UUID,
  deleted_at TIMESTAMPTZ,
  deleted_by UUID,
```

```
row_version BIGINT NOT NULL DEFAULT ۱
);
CREATE INDEX idx_user_credentials_user ON user_credentials(user_id) WHERE deleted_at IS NULL;
```

-- ۳. user_profiles

```
CREATE TABLE user_profiles (
  profile_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  user_id UUID NOT NULL REFERENCES users(user_id) ON DELETE RESTRICT, -- تغییر به RESTRICT
  national_id VARCHAR(۵۰),
  birth_date DATE,
  avatar_url VARCHAR(۵۰۰),
  gender SMALLINT,
  address TEXT,
  phone VARCHAR(۵۰),
  description VARCHAR(۵۰۰),
  created_at TIMESTAMPTZ NOT NULL DEFAULT NOW(),
  created_by UUID,
  updated_at TIMESTAMPTZ,
  updated_by UUID,
  deleted_at TIMESTAMPTZ,
  deleted_by UUID,
  row_version BIGINT NOT NULL DEFAULT ۱
);
CREATE INDEX idx_user_profiles_user ON user_profiles(user_id) WHERE deleted_at IS NULL;
```

-- ۴. tenant_users

جهت جلوگیری از نشت داده tenant_id انتقال فیلدهای هویتی و ترکیب یکتا با --

```
CREATE TABLE tenant_users (
  tenant_user_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  tenant_id UUID NOT NULL, -- (فیزیکی بر اساس اصول مرزبندی FK بدون) ارجاع منطقی به لایه ساس ادمین --
  user_id UUID NOT NULL REFERENCES users(user_id) ON DELETE RESTRICT,
  username VARCHAR(۱۰۰) NOT NULL,
  tenant_user_email VARCHAR(۱۵۰) NOT NULL,
  user_type SMALLINT NOT NULL DEFAULT ۱,
  status SMALLINT NOT NULL DEFAULT ۱,
  created_at TIMESTAMPTZ NOT NULL DEFAULT NOW(),
  created_by UUID,
  updated_at TIMESTAMPTZ,
  updated_by UUID,
  deleted_at TIMESTAMPTZ,
```

```
deleted_by UUID,  
row_version BIGINT NOT NULL DEFAULT ۱  
);
```

-- تضمین ۱۰۰٪ امنیت مالتی-تننتی: یک نام کاربری یا ایمیل فقط در سطح همان شرکت یکتاست --

```
CREATE UNIQUE INDEX uq_tenant_users_username ON tenant_users(tenant_id, username) WHERE deleted_at IS NULL;  
CREATE UNIQUE INDEX uq_tenant_users_email ON tenant_users(tenant_id, tenant_user_email) WHERE deleted_at IS NULL;  
CREATE INDEX idx_tenant_users_composite ON tenant_users(tenant_id, user_id) WHERE deleted_at IS NULL;
```

-- ۵. roles

```
CREATE TABLE roles (  
  role_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
  tenant_id UUID NOT NULL, -- برای اجباری بودن بافتار تننت NOT NULL تغییر به --  
  name VARCHAR(۱۰۰) NOT NULL,  
  code VARCHAR(۱۰۰) NOT NULL,  
  description VARCHAR(۵۰۰),  
  role_type SMALLINT NOT NULL DEFAULT ۱,  
  role_scope_type SMALLINT NOT NULL DEFAULT ۱,  
  is_system_role BOOLEAN NOT NULL DEFAULT FALSE,  
  status SMALLINT NOT NULL DEFAULT ۱,  
  created_at TIMESTAMPTZ NOT NULL DEFAULT NOW(),  
  created_by UUID,  
  updated_at TIMESTAMPTZ,  
  updated_by UUID,  
  deleted_at TIMESTAMPTZ,  
  deleted_by UUID,  
  row_version BIGINT NOT NULL DEFAULT ۱  
);  
  
CREATE UNIQUE INDEX uq_tenant_role_code ON roles(tenant_id, code) WHERE deleted_at IS NULL;
```

-- ۶. permissions

```
CREATE TABLE permissions (  
  permission_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
  code VARCHAR(۱۵۰) NOT NULL,  
  name VARCHAR(۲۰۰) NOT NULL,  
  module_name VARCHAR(۱۰۰) NOT NULL,  
  group_name VARCHAR(۱۰۰),  
  description VARCHAR(۵۰۰),  
  permission_type SMALLINT NOT NULL DEFAULT ۱,  
  status SMALLINT NOT NULL DEFAULT ۱,  
  created_at TIMESTAMPTZ NOT NULL DEFAULT NOW(),
```

```

created_by UUID,
updated_at TIMESTAMPTZ,
updated_by UUID,
deleted_at TIMESTAMPTZ,
deleted_by UUID,
row_version BIGINT NOT NULL DEFAULT 1
);
CREATE UNIQUE INDEX uq_permissions_code ON permissions(code) WHERE deleted_at IS NULL;

```

```

-- V. role_permissions
CREATE TABLE role_permissions (
role_permission_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
role_id UUID NOT NULL REFERENCES roles(role_id) ON DELETE RESTRICT,
permission_id UUID NOT NULL REFERENCES permissions(permission_id) ON DELETE RESTRICT,
access_level SMALLINT NOT NULL DEFAULT 1,
status SMALLINT NOT NULL DEFAULT 1,
created_at TIMESTAMPTZ NOT NULL DEFAULT NOW(),
created_by UUID,
updated_at TIMESTAMPTZ,
updated_by UUID,
deleted_at TIMESTAMPTZ,
deleted_by UUID,
row_version BIGINT NOT NULL DEFAULT 1
);
CREATE UNIQUE INDEX uq_role_permission ON role_permissions(role_id, permission_id) WHERE deleted_at IS NULL;

```

```

-- A. user_roles
CREATE TABLE user_roles (
user_role_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
tenant_user_id UUID NOT NULL REFERENCES tenant_users(tenant_user_id) ON DELETE RESTRICT,
role_id UUID NOT NULL REFERENCES roles(role_id) ON DELETE RESTRICT,
assigned_by UUID,
start_date TIMESTAMPTZ,
end_date TIMESTAMPTZ,
status SMALLINT NOT NULL DEFAULT 1,
created_at TIMESTAMPTZ NOT NULL DEFAULT NOW(),
created_by UUID,
updated_at TIMESTAMPTZ,
updated_by UUID,
deleted_at TIMESTAMPTZ,

```

```

deleted_by UUID,
row_version BIGINT NOT NULL DEFAULT ۱
);
CREATE UNIQUE INDEX uq_user_role ON user_roles(tenant_user_id, role_id) WHERE deleted_at IS NULL;
-- بهبود کارایی: شاخص سرچ سریع نقش‌ها برای موتور احراز دسترسی
CREATE INDEX idx_user_roles_role_lookup ON user_roles(role_id) WHERE deleted_at IS NULL;

-- ۹. scopes
CREATE TABLE scopes (
scope_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
tenant_id UUID NOT NULL, -- بر اساس اصول امنیت چندمستأجری NOT NULL تغییر به
scope_type SMALLINT NOT NULL,
reference_type VARCHAR(۱۰۰) NOT NULL,
reference_id UUID NOT NULL,
name VARCHAR(۲۰۰) NOT NULL,
description VARCHAR(۵۰۰),
status SMALLINT NOT NULL DEFAULT ۱,
created_at TIMESTAMPTZ NOT NULL DEFAULT NOW(),
created_by UUID,
updated_at TIMESTAMPTZ,
updated_by UUID,
deleted_at TIMESTAMPTZ,
deleted_by UUID,
row_version BIGINT NOT NULL DEFAULT ۱
);
CREATE INDEX idx_scopes_tenant ON scopes(tenant_id) WHERE deleted_at IS NULL;
-- برای لایه کنترل محدوده داده در بک‌آند (Polymorphic) بهبود کارایی: ایندکس ترکیبی چندشکلی
CREATE INDEX idx_scopes_polymorphic ON scopes(reference_type, reference_id) WHERE deleted_at IS NULL;

-- ۱۰. user_scope_assignments
CREATE TABLE user_scope_assignments (
user_scope_assignment_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
tenant_user_id UUID NOT NULL REFERENCES tenant_users(tenant_user_id) ON DELETE RESTRICT,
scope_id UUID NOT NULL REFERENCES scopes(scope_id) ON DELETE RESTRICT,
is_primary BOOLEAN NOT NULL DEFAULT FALSE,
start_date TIMESTAMPTZ,
end_date TIMESTAMPTZ,
status SMALLINT NOT NULL DEFAULT ۱,
created_at TIMESTAMPTZ NOT NULL DEFAULT NOW(),
created_by UUID,

```

```

updated_at TIMESTAMPTZ,
updated_by UUID,
deleted_at TIMESTAMPTZ,
deleted_by UUID,
row_version BIGINT NOT NULL DEFAULT 1
);
CREATE UNIQUE INDEX uq_user_scope ON user_scope_assignments(tenant_user_id, scope_id) WHERE deleted_at IS NULL;

```

```

-- =====
-- Layer ۴: ERP Core Identity Organization Layer
-- Tenant Level RBAC Security Model
-- =====

```

```

-- ۱. tenant_roles
-- نقش‌های کاربران داخل هر Tenant
CREATE TABLE tenant_roles (
    tenant_role_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),

    tenant_id UUID NOT NULL,

    code VARCHAR(100) NOT NULL,
    name VARCHAR(150) NOT NULL,
    description VARCHAR(500),

    status SMALLINT NOT NULL DEFAULT 1,

    created_at TIMESTAMPTZ NOT NULL DEFAULT NOW(),
    created_by UUID,

    updated_at TIMESTAMPTZ,
    updated_by UUID,

    deleted_at TIMESTAMPTZ,
    deleted_by UUID,

    row_version BIGINT NOT NULL DEFAULT 1
);

```

```
CREATE UNIQUE INDEX uq_tenant_roles_code
ON tenant_roles(tenant_id, code)
WHERE deleted_at IS NULL;
```

```
CREATE INDEX idx_tenant_roles_tenant
ON tenant_roles(tenant_id)
WHERE deleted_at IS NULL;
```

```
-- =====
-- ۲. tenant_permissions
-- مجوزهای دسترسی سطح Tenant
-- =====
```

```
CREATE TABLE tenant_permissions (
    tenant_permission_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),

    tenant_id UUID NOT NULL,

    code VARCHAR(۱۵۰) NOT NULL,
    name VARCHAR(۲۰۰) NOT NULL,

    module_name VARCHAR(۱۰۰),
    action_type VARCHAR(۵۰),
    -- CREATE
    -- READ
    -- UPDATE
    -- DELETE
    -- APPROVE
    -- EXECUTE

    description VARCHAR(۵۰۰),

    status SMALLINT NOT NULL DEFAULT ۱,

    created_at TIMESTAMPTZ NOT NULL DEFAULT NOW(),
    created_by UUID,
```

```

        updated_at TIMESTAMPTZ,
        updated_by UUID,

        deleted_at TIMESTAMPTZ,
        deleted_by UUID,

        row_version BIGINT NOT NULL DEFAULT 1
    );

```

```

CREATE UNIQUE INDEX uq_tenant_permissions_code
ON tenant_permissions(tenant_id, code)
WHERE deleted_at IS NULL;

```

```

CREATE INDEX idx_tenant_permissions_module
ON tenant_permissions(tenant_id, module_name)
WHERE deleted_at IS NULL;

```

```

-- =====
-- ۳. tenant_user_roles
-- به نقش‌ها Tenant اتصال کاربران
-- =====

```

```

CREATE TABLE tenant_user_roles (
    tenant_user_role_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),

    tenant_id UUID NOT NULL,

    user_id UUID NOT NULL,
    -- Logical Reference to Layer ۴ Users

    tenant_role_id UUID NOT NULL
        REFERENCES tenant_roles(tenant_role_id)
        ON DELETE RESTRICT,

    created_at TIMESTAMPTZ NOT NULL DEFAULT NOW(),

```



```
    created_by UUID,

    updated_at TIMESTAMPTZ,
    updated_by UUID,

    deleted_at TIMESTAMPTZ,
    deleted_by UUID,

    row_version BIGINT NOT NULL DEFAULT 1
);
```

```
CREATE UNIQUE INDEX uq_tenant_user_roles
ON tenant_user_roles(
    tenant_id,
    user_id,
    tenant_role_id
)
WHERE deleted_at IS NULL;
```

```
CREATE INDEX idx_tenant_user_roles_user
ON tenant_user_roles(
    tenant_id,
    user_id
)
WHERE deleted_at IS NULL;
```

```
-- =====
-- ۴. tenant_role_permissions
-- اتصال نقش‌ها به مجوزها
-- =====
```

```
CREATE TABLE tenant_role_permissions (
    tenant_role_permission_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),

    tenant_id UUID NOT NULL,
```

```
tenant_role_id UUID NOT NULL
    REFERENCES tenant_roles(tenant_role_id)
    ON DELETE RESTRICT,
```

```
tenant_permission_id UUID NOT NULL
    REFERENCES tenant_permissions(tenant_permission_id)
    ON DELETE RESTRICT,
```

```
created_at TIMESTAMPTZ NOT NULL DEFAULT NOW(),
created_by UUID,
```

```
updated_at TIMESTAMPTZ,
updated_by UUID,
```

```
deleted_at TIMESTAMPTZ,
deleted_by UUID,
```

```
row_version BIGINT NOT NULL DEFAULT 1
);
```

```
CREATE UNIQUE INDEX uq_tenant_role_permissions
ON tenant_role_permissions(
    tenant_id,
    tenant_role_id,
    tenant_permission_id
)
WHERE deleted_at IS NULL;
```

```
CREATE INDEX idx_tenant_role_permissions_role
ON tenant_role_permissions(
    tenant_role_id
)
WHERE deleted_at IS NULL;
```